Source code:

```
import pandas as pd import
numpy as np import
matplotlib.pyplot as plt import os
import glob
# Search for Life Expectancy Dataset files =
glob.glob("/drive/notebooks/**/*.csv", recursive=True) files +=
glob.glob("*.csv") files = list(set(files))
df = None
for file in files:     try:
        temp = pd.read_csv(file)         temp.columns
= temp.columns.str.strip()
        if any("life expectancy" in c.lower() for c in temp.columns):
            df = temp             print("Dataset
loaded:", file)             break
    except Exception:
        continue
# Use sample data if dataset is unavailable if df
is None:
                print("Dataset not found. Using sample data for demonstration.")
    df = pd.DataFrame({
        "Country": ["A", "B", "C", "D", "E", "F", "G", "H"],
       "Life expectancy": [72, 65, 78, 60, 75, 68, 80, 63],
        "GDP": [5000, 2000, 9000, 1500, 7000, 3500, 11000, 1800],
        "Schooling": [12, 8, 15, 6, 14, 10, 16, 7],
        "Adult Mortality": [150, 250, 100, 300, 120, 200, 80, 280],
        "Infant Mortality": [20, 45, 10, 60, 15, 35, 8, 50]
    })
```

```
target = next(
    (c for c in df.columns if c.lower().strip() == "life expectancy"),
    None
)

if target is None:
    print("Life expectancy column is unavailable.") else:
    df[target] = pd.to_numeric(df[target], errors="coerce")     df =
df.dropna(subset=[target])

    print("\nDataset Head:")     print(df.head())

    print("\nDataset Information:")
    df.info()

    print("\nStatistical Summary:")
   print(df.describe())

    print("\nAverage Life Expectancy:", round(df[target].mean(), 2))
print("Maximum Life Expectancy:", round(df[target].max(), 2))
print("Minimum Life Expectancy:", round(df[target].min(), 2))

    # Histogram     plt.figure(figsize=(7, 4))
plt.hist(df[target], bins=10, edgecolor="black")
plt.title("Life Expectancy Distribution")
plt.xlabel("Life Expectancy")
 plt.ylabel("Frequency")
 plt.tight_layout()     plt.show()
```

```
    # Correlation analysis      numeric_df =
df.select_dtypes(include=np.number)      corr =
numeric_df.corr()[target].drop(target).dropna()

    if not corr.empty:
        print("\nMajor Factors (Correlation):")
print(corr.sort_values(ascending=False))

        corr.sort_values().plot(kind="bar", figsize=(8, 4))
plt.title("Factors Associated with Life Expectancy")
plt.xlabel("Factors")
 plt.ylabel("Correlation")
 plt.tight_layout()
       plt.show()
    print("\nAnalysis completed successfully.")
```

Output:

Dataset not found. Using sample data for demonstration.

Dataset Head:

	Country	Life expectancy	GDP	Schooling	Adult Mortality	Infant Mortality
0	A	72	5000	12	150	20
1	B	65	2000	8	250	45
2	C	78	9000	15	100	10
3	D	60	1500	6	300	60

```
4       E              75  7000        14             120              15

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 6 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Country           8 non-null      str
 1   Life expectancy   8 non-null      int64
 2   GDP               8 non-null      int64
 3   Schooling         8 non-null      int64
 4   Adult Mortality   8 non-null      int64
 5   Infant Mortality  8 non-null      int64
dtypes: int64(5), str(1)
memory usage: 424.0 bytes

Statistical Summary:
       Life expectancy          GDP  Schooling  Adult Mortality  \
count         8.000000      8.000000   8.000000         8.000000
mean         70.125000   5100.000000  11.000000       185.000000
std           7.278883   3578.906457   3.817254        84.852814
min          60.000000   1500.000000   6.000000        80.000000
25%          64.500000   1950.000000   7.750000       115.000000
50%          70.000000   4250.000000  11.000000       175.000000
75%          75.750000   7500.000000  14.250000       257.500000
max          80.000000  11000.000000  16.000000       300.000000

       Infant Mortality
```

count 8.000000

mean 30.375000

std 19.848983

min 8.000000

25% 13.750000

50% 27.500000

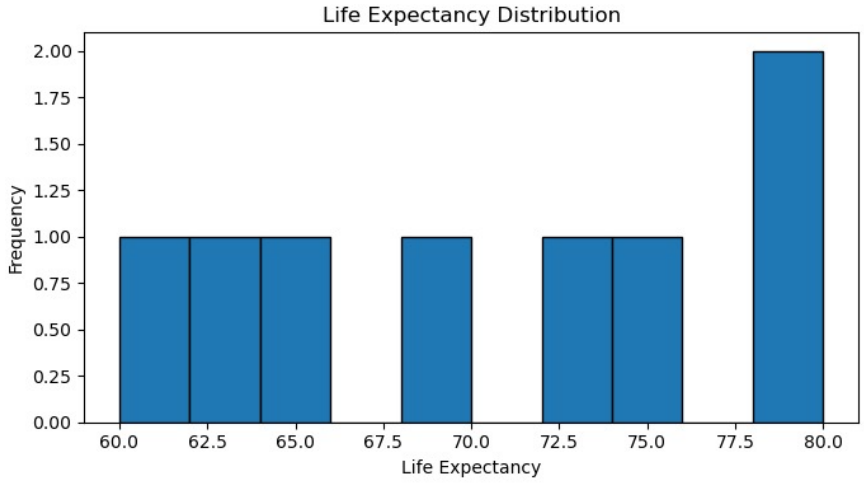
75% 46.250000

max 60.000000

Average Life Expectancy: 70.12

Maximum Life Expectancy: 80

Minimum Life Expectancy: 60



Major Factors (Correlation):

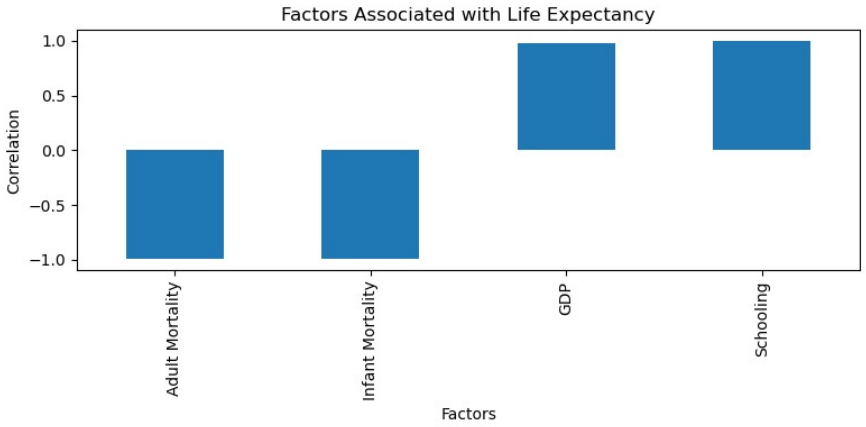
Schooling 0.997442

GDP 0.972289

Infant Mortality -0.989149

Adult Mortality -0.993423

Name: Life expectancy, dtype: float64


Factors Associated with Life Expectancy
Correlation
1.0
0.5
0.0
−0.5
−1.0
Adult Mortality
Infant Mortality
GDP
Schooling
Factors